Progressive Education Society's
Modern College of Engineering, Pune-05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Department of S.Y. B. Tech [AIDS]

Assignment No: 06

SUBJECT: Java Programming

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Title / Problem Statement :

Write a program to develop a simple banking system where an account is created with a minimum balance of ₹1000. The system should allow depositing and withdrawing amounts. If a withdrawal causes the balance to go below ₹1000, the program should throw a custom exception named lessbalanceexception, which displays a message: "withdraw amount (rs) is not valid. insufficient balance." Additionally, demonstrate the use of a built-in exception (arithmeticexception) by performing a division operation where the divisor is zero, and handle the exception.

Input :

```java
import java.util.Scanner;


// Custom Exception
class LessBalanceException extends Exception {
    LessBalanceException(double amount) {
        super("Withdraw amount (Rs " + amount + ") is not valid. Insufficient
balance.");
    }
}

// Account Class
class Account {
    double balance;

    // Constructor
    Account(double balance) {
        if (balance < 1000) {
            this.balance = 1000;
            System.out.println("Minimum balance set to Rs 1000");
        } else {
            this.balance = balance;
        }
    }

    // Deposit Method
    void deposit(double amount) {
        balance += amount;
        System.out.println("Deposited: Rs " + amount);
    }

    // Withdraw Method
    void withdraw(double amount) throws LessBalanceException {
        if ((balance - amount) < 1000) {
            throw new LessBalanceException(amount);
        } else {
            balance -= amount;
            System.out.println("Withdrawn: Rs " + amount);
        }
    }
```

```java
    void display() {
        System.out.println("Current Balance: Rs " + balance);
    }
}

// Main Class
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter initial balance: ");
            double bal = sc.nextDouble();

            Account acc = new Account(bal);

            System.out.print("Enter deposit amount: ");
            double dep = sc.nextDouble();
            acc.deposit(dep);

            System.out.print("Enter withdrawal amount: ");
            double wd = sc.nextDouble();

            try {
                acc.withdraw(wd);
            } catch (LessBalanceException e) {
                System.out.println(e.getMessage());
            }

            acc.display();

            // Demonstrating ArithmeticException
            System.out.println("\nDemonstrating ArithmeticException:");
            int a = 10, b = 0;
            int result = a / b;  // will cause exception
            System.out.println("Result: " + result);

        } catch (ArithmeticException e) {
            System.out.println("Error: Cannot divide by zero!");
```

```
        } catch (Exception e) {
            System.out.println("Error: Invalid input!");
        }

        sc.close();
    }
}
```

Output :

```
class Account {
    void display() {
    }
}

// Main Class
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print("Enter initial balance: ");
            double bal = sc.nextDouble();

            Account acc = new Account(bal);

            System.out.print("Enter deposit amount: ");
            double dep = sc.nextDouble();
            acc.deposit(dep);

            System.out.print("Enter withdrawal amount: ");
            double wd = sc.nextDouble();
```

```
Enter initial balance: 2000
Enter deposit amount: 500
Deposited: Rs 500.0
Enter withdrawal amount: 800
Withdrawn: Rs 800.0
Current Balance: Rs 1700.0

Demonstrating ArithmeticException:
Error: Cannot divide by zero!
prasad_mali07@ASUS-Vivobook-16-M1607KA:/media/prasad_mali07/408423E38423DA64/College/College Coding/Java/assignment_06$
```